



Date: 04/12/2025

Lab Practical #1:

- 1) Write a C program to count the number of 1's and 0's in the binary representation of a number.
- 2) Write a C program to perform bitwise AND, OR, XOR, and NOT operations on integers.
- 3) Write a C program that takes a string and performs bitwise AND and XOR operations with a constant value (e.g., 127). (Simulates simple encryption idea)
- 4) Write a C program to implement left and right circular bit rotations.
- 5) Write a C program to swap two variables without using a third variable (using XOR).

Program-1:

```
#include<stdio.h>

int main () {
    printf("Enter n : ");
    int n, os=0, zs=0, rem=0;
    scanf("%d", &n);
    while (n != 0) {
        if (n & 1) os++;
        else zs++;
        n /= 2;
    }
    printf("Zeros %d, Ones %d\n", zs, os);
    return 0;
}
```

Output:

```
gcc 1.c && ./a.out
Enter n : 5
Zeros 1, Ones 2
```



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Program-2:

```
#include <stdio.h>

int main() {
    int x, y;

    printf("Enter two integers: ");
    scanf("%d %d", &x, &y);

    printf("Bitwise AND: %d & %d = %d\n", x, y, x & y);
    printf("Bitwise OR: %d | %d = %d\n", x, y, x | y);
    printf("Bitwise XOR: %d ^ %d = %d\n", x, y, x ^ y);
    printf("Bitwise NOT: ~%d = %d\n", x, ~x);

    return 0;
}
```

Output:

```
gcc 2.c && ./a.out
Enter two integers: 10
20
Bitwise AND: 10 & 20 = 0
Bitwise OR: 10 | 20 = 30
Bitwise XOR: 10 ^ 20 = 30
Bitwise NOT: ~10 = -11
```



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Program-3:

```
#include <stdio.h>

int main() {
    char s[100];
    int k = 127;

    printf("Enter a string: ");
    scanf("%s", s);

    printf("Original: %s\n", s);

    for(int i = 0; s[i] != '\0'; i++) {
        printf("Char '%c': AND=%d, XOR=%d\n", s[i], s[i] & k, s[i] ^ k);
    }

    return 0;
}
```

Output:

```
gcc 3.c && ./a.out
Enter a string: soham
Original: soham
Char 's': AND=115, XOR=12
Char 'o': AND=111, XOR=16
Char 'h': AND=104, XOR=23
Char 'a': AND=97, XOR=30
Char 'm': AND=109, XOR=18
```

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Program-4:

```
#include <stdio.h>

int main() {
    int n, r, d;

    printf("Enter a number: ");
    scanf("%d", &n);

    printf("Enter number of rotations: ");
    scanf("%d", &r);

    printf("Enter direction (1=left, 2=right): ");
    scanf("%d", &d);

    r %= 32;

    if(d == 1) {
        printf("Left circular rotation: %d\n", (n << r) | (n >> (32 - r)));
    } else {
        printf("Right circular rotation: %d\n", (n >> r) | (n << (32 - r)));
    }

    return 0;
}
```

Output:

```
~/Desktop/sem-6/ins/lab-1 (5.793s)
gcc 4.c && ./a.out
Enter a number: 12
Enter number of rotations: 2
Enter direction (1=left, 2=right): 1
Left circular rotation: 48

~/Desktop/sem-6/ins/lab-1 (6.006s)
gcc 4.c && ./a.out
Enter a number: 12
Enter number of rotations: 2
Enter direction (1=left, 2=right): 2
Right circular rotation: 3
```



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Program-5:

```
#include <stdio.h>

int main() {
    int x, y;

    printf("Enter two integers to swap: ");
    scanf("%d %d", &x, &y);

    printf("Before swap: x = %d, y = %d\n", x, y);

    x = x ^ y;
    y = x ^ y;
    x = x ^ y;

    printf("After swap: x = %d, y = %d\n", x, y);

    return 0;
}
```

Output:

```
~/Desktop/sem-6/ins/lab-1 (2.202s)
gcc 5.c && ./a.out
Enter two integers to swap: 10
20
Before swap: x = 10, y = 20
After swap: x = 20, y = 10
```